

Product name

COSMOS FTP/TELNET CLIENTS

Product number

211154E

DESCRIPTION OF PRODUCT

This product allows Terminal emulation and file access between an ND machine and remote systems running TCP/IP protocols over Ethernet.

This product can run over COSMOS TCP/IP Gateway in the Ethernet II controller and over TCP/IP Basic Module for ND-5000 in the Ethernet III controller.

The Telnet Client in COSMOS FTP/Telnet Clients is compatible with Telnetd in Berkeley Unix 4.3. It is not compatible with Telnetd in Berkeley Unix 4.2.

A Domain Name Service (DNS) Resolver is included in these clients. This allows you to access a Domain Name Server when looking up IP-adresses (instead of using the local (SYSTEM)AIP-HOSTS:SYMB file). If the file (SYSTEM)AIP-RESOLVER:SYMB does not exist, or contains no 'nameserver' entries, the Resolver code will not be used, and the client will then access the (SYSTEM)AIP-HOSTS:SYMB file just like previous versions. However, if a 'nameserver' entry does exist, a name-to-address resolution request will be sent to the DNS Server at the IP-address specified in the 'nameserver' entry.

REASON(S) FOR NEW PRODUCT OR VERSION

Version E02

- The FTP client is now delivered in both 100 and 500(0) versions.
- The FTP client now allows files with "holes", such as :DOM or 2-bank :PROG files, to be transferred.
- The RSH client delivered with version D01 of this product has been removed. The new Sintran III Remote Shell Utility product contains a similar client.
- See page 7 and onwards for a list of other modifications and error corrections.

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1 PREREQUISITES

The following things are required in order to use this product.

1.1 HARDWARE PREREQUISITES

The following hardware must be present for this product to be used:

CPU type (any of the following):

- 100 CX
- 110 CX
- 120 CX
- 500
- 5000

Floating format: Any

Other hardware:

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None

1.2 SOFTWARE PREREQUISITES

The following software must be installed on the system for this product to be used:

Operating system:

Name

Version/Revision

SINTRAN III L05 PatchFile >= 2000B

Other software:

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TCP/IP Basic Module III.....

211327 >= D00

OR

-

COSMOS TCP/IP Gateway for Ethernet..

211185 >= D01

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1.3 OTHER PREREQUISITES

The following are necessary for this product to be used.

Minimum Permanent Mass Storage:

User	Space (pages)	Number of files
------	---------------	-----------------

Any User.....	250	2
---------------	-----	---

	Minimum	Maximum
Number of segments (ND-100)	0	0

Number of RT descriptions	0	0
---------------------------	---	---

Space required on segment files: 0 pages

Requirements regarding internal devices:

None.

Requirements regarding batch processors:

None.

2 DEPENDENCIES

In addition to the above listed hardware and/or software prerequisites, the following dependencies exist between this and other products. The hardware and software listed here may, for example, be necessary to use the product's full functionality.

2.1 HARDWARE DEPENDENCIES

There are dependency relationships between this product and the following hardware:

None.

2.2 SOFTWARE DEPENDENCIES

There are dependency relationships between this product and the following software:

None.

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2.3 OTHER DEPENDENCIES

In addition to the hardware and software dependency relationships listed above, you should be aware of the following dependencies:

To utilize the DNS Resolver at least one Domain Name Server must be available on the net.

3 DOCUMENTATION

Optional (must be purchased)

TitleNumber

OpenLAN Network Supervisor Guide.....

ND830107EN 3

OpenLAN DNS Operation Guide.....

ND830145EN 1

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4 INSTALLATION PROCEDURE

4.1 AUTOMATIC INSTALLATION

None.

4.2 NON-AUTOMATIC INSTALLATION

Insert the floppy and enter directory:

```
@ENTER-DIRECTORY 211154E-XX-01D,floppy-disc-<no.>,<unit>
```

Copy the files TELNET-CLIEN-E:PROG and either FTP-CLIENT-E:PROG or FTP-CLIENT-E:DOM to a user of your choice with public read access, and dump them reentrant:

```
@DUMP-PROGRAM-REENT TELNET-CLIENT (<USERNAME>)TELNET-CLIEN-E:PROG
and
@DUMP-PROGRAM-REENT FTP-CLIENT    (<USERNAME>)FTP-CLIENT-E:PROG
or
@ND
DEFINE-STANDARD-DOM FTP-CLIENT    (<USERNAME>)FTP-CLIENT-E:DOM
EXIT
```

IF name-to-address resolution is to be performed by a domain name server, copy the AIP-RESOLVER:SYMB file to user SYSTEM with public read access, and edit it to include the local default 'DOMAIN' name (to be added to all non-fully-qualified domain names), and the IP-address of at least one 'NAMESERVER' on the net.

If TCP/IP is on a remote computer, copy the files: AIP-HOSTS:SYMB, AIP-SERVICES:SYMB, AIP-NETWORKS:SYMB, and AIP-PROTOCOL:SYMB to user SYSTEM with public read access and then define *TCPGATE in XMSG to where the servers are installed. (Note that TCP/IP can only be on a remote computer if you are using the COSMOS TCP/IP Gateway product).

Add your own internet-address to the AIP-HOSTS:SYMB file using an editor.

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5 MODIFICATIONS

5.1 NEW FUNCTIONS/COMMANDS

- The FTP client is now delivered as both a ND-100 :PROG and a ND-500 :DOM file. You may choose to install either one (or both).
- The FTP client now supports transfer Mode COMPRESS in addition to STREAM. Compress is particularly well suited for sending binary files over slow, X.25-based IP-routers.
- The FTP client now supports transfer Struct's PAGE and RECORD in addition to FILE. Page is useful when transferring binary files with "holes" between two Sintran systems. Record is best suited for fixed-length record-structured files.
- The FTP client now supports transfer Type LOCAL 32/16 in addition to ASCII, BINARY (ie. Image) and TENEX (ie. Local 8). Type Local 32 is useful, together with Struct Page (and Mode Compress), when transferring binary files with "holes" between two (possibly distant) Sintran systems. Note that Page/Local 32 effectively leads to a destination file that is an exact replica of the source, while File/Binary (which must be used if the client is a non-Sintran system) leads to a destination file where pages missing in the source file have been completely null-expanded.
- The new FTP client command CONV toggles local file name conversion according to the file-naming syntax used by the current SITE type as set. Note that this is a quite different functionality from that given by the Convert-Filename command present in previous versions of this product. (See SITE below.)
- The new FTP client command SITE enables you to inspect the remote servers site-dependant variables and current server session status, provided the server supports the FTP SITE and STAT commands; and/or force the client to accept and use the remote file-naming syntax of the server, provided this is either a Unix, MS-DOS or VMS site. The command "Site Unix,yes" will thus give you approximately the same functionality as the "Convert-Filename" command present in previous versions of this product.

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5.2 CHANGED FUNCTIONS/COMMANDS

- The FTP client command LCD now allows you to change local working directory (ie. Sintran user-area): file-access will equal that of the user-area current when entering the FTP-client, called home directory. On exit you will be back at the home directory, and all Sintran commands done from the client will by default work towards the home directory.
- The FTP client commands MGET and MPUT now allows the user to modify each destination file name proposed by the client, provided that PROMPTing is turned on.
- Refer to the Cosmos Telnet/FTP Client User Guide for minor changes to some other FTP client commands.

5.3 REMOVED FUNCTIONS/COMMANDS

- The RSH client delivered with version D01 of this product has been removed. A similar client is included with the new Sintran III Remote Shell Utility product.
- The FTP client command "Convert-Filename", present in previous versions of this product, has been removed. Use The SITE command (see "New functions/commands" above) to achieve approximately the same functionality.
- The FTP client commands "Form", "MLS" and "MDIR", dummy in previous versions, has been removed.

5.4 OTHER MODIFICATIONS

- The Telnet client can now be started on any Tcp-device, provided connection to the server is done on the command line, and the optional <tcp-device> parameter is specified:
@telnet-client server-host-name <tcp-device <remote-port>>
- The Sintran/Unix file-name conversion rules used by the FTP client has been slightly changed. The most noteworthy change is that the client will limit any proposed unix file-name to 14 characters. All "cuts" needed to make sintran files fit into this scheme, are done on the file-name part, while file-type remains un-abridged.

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6 ERRORS CORRECTED

- When in local echo modus, the Telnet client will now process Line-feed after a received Carriage-return correctly.
- The Telnet client will now handle Wont-option processing correct.
- The Telnet client will show the server's IP-address at connect if the server's host or domain name is unknown.

7 ERRORS KNOWN BUT NOT CORRECTED

None.